KT1

Ruhr-Universität Bochum
Experimental Physics I

WINTER 2026/27 COURSE 160202

Introduction to Nuclear and Particle Physics

Exercise sheet 3

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Distributed	November 3, 2026
Due	November 10, in the lecture

Homework – problems from Thomson

Textbook. Mark Thomson, *Modern Particle Physics*, Cambridge University Press, first edition (2013).

Use the problem statements and figures in your copy of the textbook. Show your reasoning, state any assumptions, and include units in numerical answers.

Exercise 1

Thomson 1.7

Energy loss of charged particles: $\mathrm{d}E/\mathrm{d}x$.

Exercise 2

Thomson 3.6

Luminosity.

Exercise 3

Thomson 3.8

Flux.

Submission

Write your solutions by hand and work individually. Hand in your solutions on **November 10, 2026, at the lecture**. AI assistance is permitted for preparation; be ready to explain your reasoning independently.

Sheet 3 – Class work 6 November 2026

1. Two-body phase space and the Dalitz plot

Use natural units ($c = \hbar = 1$) and the convention

$$\mathrm{d}\Phi_n(P) = (2\pi)^4 \delta^{(4)}\left(P - \sum_i p_i\right) \prod_{i=1}^{n} \frac{\mathrm{d}^3\mathbf{p}_i}{(2\pi)^3\, 2E_i}.$$

(a) For $P \to p_1 + p_2$ with $P^2 = M^2$, work in the parent rest frame. Use the three-momentum delta function, then the energy delta function with its Jacobian, to derive $\mathrm{d}\Phi_2/\mathrm{d}\Omega$ and the integrated Φ_2. Express the result using

$$\lambda(x, y, z) = x^2 + y^2 + z^2 - 2xy - 2xz - 2yz.$$

Count the independent degrees of freedom before and after the constraints.

(b)* For $P \to p_1 + p_2 + p_3$, define

$$s_{ij} = (p_i + p_j)^2.$$

Count the remaining degrees of freedom and separate out the overall orientation. Using the factorization

$$\mathrm{d}\Phi_3 = \frac{\mathrm{d}s_{12}}{2\pi} \mathrm{d}\Phi_2(P; q, p_3)\, \mathrm{d}\Phi_2(q; p_1, p_2), \qquad q = p_1 + p_2,$$

show that, inside the kinematic boundary,

$$\frac{\mathrm{d}\Phi_3}{\mathrm{d}s_{12}\, \mathrm{d}s_{23}} = \text{const.}$$

Explain why a constant (spin-summed/averaged) matrix element and uniform acceptance imply

$$\frac{\mathrm{d}N}{\mathrm{d}s_{12}\, \mathrm{d}s_{23}} = \text{const.}$$

2. Threshold energies – Griffiths, Problem 3.15

For a stationary target proton, find the projectile threshold *total energy* and *kinetic energy* for each reaction:

(a) $p + p \to p + p + \pi^0$, (b) $p + p \to p + p + \pi^+ + \pi^-$,
(c) $\pi^- + p \to p + \bar{p} + n$, (d) $\pi^- + p \to K^0 + \Sigma^0$,
(e) $p + p \to p + \Sigma^+ + K^0$.

Derive the threshold condition from

$$s = (p_a + p_b)^2$$

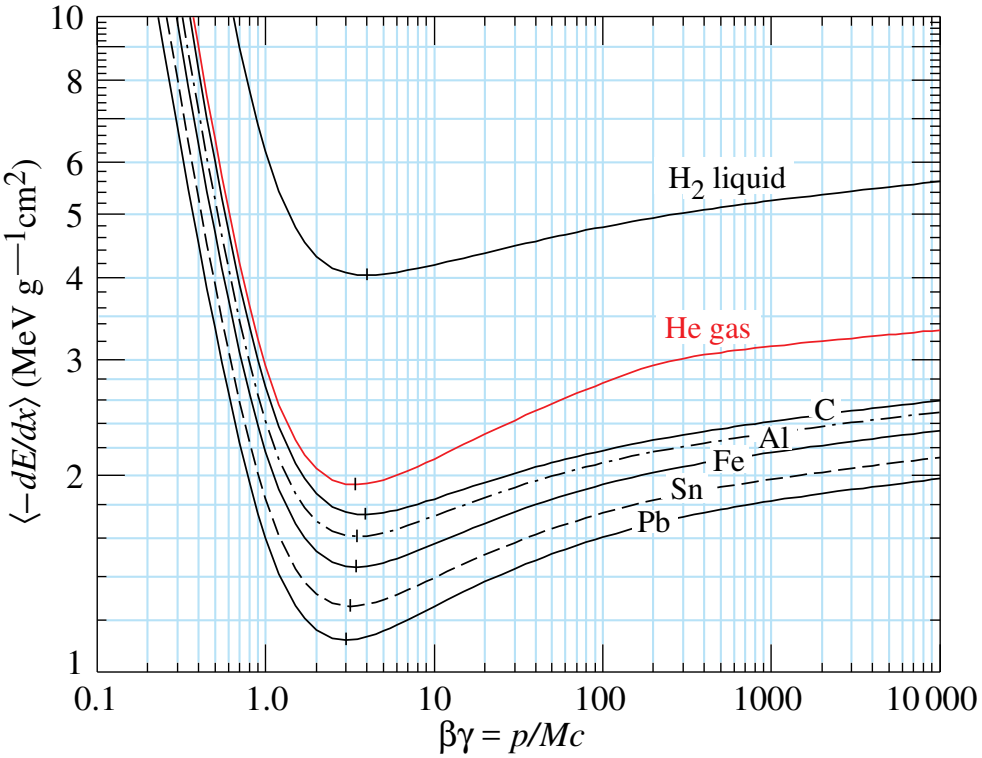
and the minimum final-state centre-of-mass energy. Use masses (in MeV/c^2):

$$m_p = 938.27, \quad m_n = 939.57, \quad m_{\pi^0} = 134.98,$$
$$m_{\pi^\pm} = 139.57, \quad m_{K^0} = 497.61, \quad m_{\Sigma^0} = 1192.64,$$
$$m_{\Sigma^+} = 1189.37.$$

3. Reading a Bethe–Bloch curve

The ordinate is the *mass stopping power*,

$$S = -\frac{1}{\rho}\left\langle \frac{\mathrm{d}E}{\mathrm{d}x} \right\rangle .$$



Source: Particle Data Group, *Passage of particles through matter*, stopping-power plot (lecture material).

(a) Read off the approximate $\beta\gamma$ and S at the minimum for aluminium. Compute the corresponding velocity v and Lorentz boost factor γ.

(b) Calculate the corresponding momenta of a muon, pion and proton. Use the masses (in MeV/c^2):

$$m_\mu = 105.66, \qquad m_\pi = 139.57, \qquad m_p = 938.27.$$

(c) Estimate the mean energy lost by a minimum-ionizing singly charged particle in 1 mm of aluminium, with density

$$\rho = 2.70\,\mathrm{g\,cm^{-3}}.$$

(d) At a momentum of 1 GeV/c, which loses more energy in aluminium: a pion or a proton? Explain using the plot.